Mintae Kim

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RESEARCH AREAS

High-level: Robotics, Reinforcement Learning, World Models, VLA, Control
Low-level: Humanoid Loco-manipulation, World Action Models, VLA Post-training, Offline GCRL

EDUCATION

UC Berkeley

Ph.D. Candidate, Mechanical Engineering (Robotics & Control) Aug 2023 – May 2028

- Affiliation: Hybrid Robotics Lab, Berkeley AI Research (BAIR), *The Henry Lurie Family Fellow*
- Minors: EECS, Mathematics
- Course Highlights: Advanced Control Systems, Nonlinear Control, Multi-Agent Systems Control, Statistical Signal Processing, Mathematical Methods for Physical Sciences, Financial Engineering Systems

M.S., Mechanical Engineering (Robotics & Control) Aug 2023 – Dec 2025

- Thesis: World Model-based Experience Transfer for Robust and Sample-efficient Reinforcement Learning

KAIST Feb 2019 – Feb 2022

B.S., Industrial and Systems Engineering, Mathematics, Computer Science (Interdisciplinary)

- Graduated in three years with full scholarship for academic excellence.
- Course Highlights: Advanced Physics, Advanced Linear Algebra, Convex Optimization*, Stochastic Processes*, Data Structure and Analysis, Statistical Machine Learning, Deep Generative Models* (*: Graduate courses)

Daejeon Dongsin Science High School Mar 2016 – Feb 2019

- *Summa Cum Laude* (1st in Physics, Mathematics, and Computer Science)

RESEARCH EXPERIENCE

Hybrid Robotics Lab Aug 2023 – May 2028

- **Humanoid Loco-manipulation:** sys 0: fast motion tracking, sys 1: trajectory generator, sys 2: VLM
- **Robust & Safe Reinforcement Learning:** self-supervised safety from sparse signals, robustness under model uncertainty and disturbances, and adaptation to non-stationary dynamics.
- **VLA, World Models, World-Action Models:** latent world model, world model-based trajectory generation and MPC, joint world-action flows, model-based planning, and offline goal-conditioned RL for VLA post-training.
- **Differentiable Optimization & Implicit Layers:** differentiable trajectory optimization, implicit differentiation, and end-to-end RL with contraction layers (for robustness) and control barrier layers (for safety).
- **Offline & Model-Based RL:** trajectory-regularized KL control, offline-to-online RL leveraging MPC, in-distribution imagination for offline RL, trajectory-regularized stochastic optimal control.

Ados at Samsung Electronics Apr 2022 – Jul 2023

- Developed **learning-based cyber-physical security systems** for industrial applications, leveraging vision-based generative models, and anomaly detection; deployed on quadrotors and smart surveillance systems.
- Built **open-set action recognition system** for surveillance **anomaly detection**, enabling detection of unseen behaviors; outperformed supervised, generative, and predictive baselines. Contract position at Samsung.

KAIST Dec 2020 – Jun 2021

- Developed **recurrent RL algorithms for optimal high-frequency trading strategies** in a limit order book.
- Developed vision-based Seoul city pedestrian safety systems.

WORK EXPERIENCE

Graduate Student Researcher

Berkeley Artificial Intelligence Research (BAIR), UC Berkeley *Aug 2023 – May 2028*

- Humanoid loco-manipulation, RL, world models, and SOC for robotics, and decision-making under uncertainty.
- Developed algorithms in robust/safe online RL, offline RL, and offline-to-online transfer.
- Proposed distributionally robust RL and safe RL methods under model uncertainty.

Graduate Student Instructor

UC Berkeley *Jan 2025 – May 2026*

- Instructed MEC237/EECS222 (Nonlinear Control; Spring 2026), MEC292B (Feedback Control of Legged Robots; Fall 2025), EECS225A (Statistical Signal Processing; Spring 2025).
- Authored MEC237/EECS222 lecture materials on RL and learning-based control.

Deep Learning Researcher

Ados at Samsung Electronics *Apr 2022 – Jul 2023*

- Developed AI-based cybersecurity systems for industrial environments.
- Built open-set action recognition models for anomaly detection in surveillance systems.

Tutor (Mathematics & Physics)

The Princeton Review *Feb 2019 – Jul 2023*

- Taught competition math (IMO, USAMO, AIME, KMO) and AP courses (Calculus, Physics).

PUBLICATIONS

Joint World Action Models through Differentiable Trajectory Optimization
Mintae Kim, Koushil Sreenath
2027 International Conference on Learning Representations (ICLR) – Under review

Learning Safety from Sparse Signals: Goal-Conditioned Reach-Avoid-Stay Reinforcement Learning with Safety Occupancy Guarantees
Mintae Kim, Koushil Sreenath
2027 International Conference on Learning Representations (ICLR) – Under review

Latent Regime World Models for Reinforcement Learning under Nonstationary Dynamics
Mintae Kim, Andrew D. Li, Koushil Sreenath
2027 IEEE L-CSS / American Control Conference (ACC) – Under review

Trajectory-Regularized Offline Reinforcement Learning via KL Control
Mintae Kim, Koushil Sreenath
2027 American Control Conference (ACC) – Under review

In-Distribution Imagination for Model Based Offline Reinforcement Learning
Mintae Kim, Koushil Sreenath
2026 Reinforcement Learning Conference (RLC)

Robust Safety in Reinforcement Learning via Implicit Barrier Layers
Mintae Kim, Koushil Sreenath
2026 Conference of Decision and Control (CDC)

Trajectory-Regularized Stochastic Optimal Control via KL Divergence
Mintae Kim, Koushil Sreenath
2026 Conference of Decision and Control (CDC)

Robust Adversarial Policy Optimization Under Dynamics Uncertainty
Mintae Kim, Koushil Sreenath
2026 Conference of Decision and Control (CDC)

WOMBET: World Model-based Experience Transfer for Robust and Sample-efficient RL
Mintae Kim, Koushil Sreenath
2026 Learning for Dynamics and Control Conference (L4DC)

Learning-based Trajectory Tracking for Bird-inspired Flapping-wing Robots
Jiaze Cai, Vishnu Sangli, Mintae Kim, Koushil, Sreenath
2025 American Control Conference (ACC)

Estimation of Aerodynamical Forces in Dynamic Morphing Wing Flight
Bibek Gupta*, Mintae Kim*, Albert Park, Eric Sihite, Alireza Ramezani, Koushil Sreenath
2025 International Conference of Intelligent Robots and Systems (IROS)

RoVerFly: Robust and Versatile Implicit Hybrid Control of Quadrotor-Payload Systems
Mintae Kim, Jiaze Cai, Koushil Sreenath
2027 American Control Conference (ACC) – Under review

Finite Memeory Belief Approximation for Optimal Control in POMDPs
Mintae Kim, Koushil Sreenath
2027 American Control Conference (ACC) – Under review

AWARDS

The Henry Lurie Family Fellowship – UC Berkeley *Best Graduate Student* Award *Jun 2026*

Berkeley Fellowship *Aug 2023*

KAIST Undergraduate Coding Competition – 2nd Place Jul 2021

KAIST Presidential Fellowship *Feb 2019 and Aug 2021*

KAIST Entrance Scholarship – 1st Place in Math and Physics *Feb 2019*

Korea Science Highschool Mathematics Competition – 1st Place (Individual) *Jun 2017*

SKILLS

Programming: Python, C++, R, Java
Deep Learning: PyTorch, Jax, Cuda
Others: Kubernetis, Claude Code, MuJoCo